

Decision Tree Report

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Abstract

For our first assignment in the Introduction to ML course we were asked to implement a decision tree algorithm to determine indoor locations based on WIFI signal strengths collected from a mobile phone. The following report presents our results along with a short analysis.

1 Evaluation

1.1 Metrics

First we created a decision tree class. Then, we were asked to train decision trees on both the clean and the noisy datasets provided to us and to analyse our results using 10-fold cross validation. The results for each of these stages are below.¹

1.1.1 Clean Dataset

The resulting metrics for the performance of our decision tree on the clean dataset are as follows:

Actual	Predicted			
	Class 1	Class 2	Class 3	Class 4
Class 1	49.6	0.0	0.2	0.2
Class 2	0.0	48.9	1.1	0.0
Class 3	0.2	1.2	48.4	0.2
Class 4	0.2	0.0	0.2	49.6

Confusion Matrix 1: Clean dataset with no pruning

Accuracy: 0.983

Precision	Class 1	Class 2	Class 3	Class 4
	0.992	0.976	0.970	0.992
Recall	Class 1	Class 2	Class 3	Class 4
	0.992	0.978	0.968	0.992
F1-score	Class 1	Class 2	Class 3	Class 4
	0.992	0.977	0.969	0.992

¹Results were averaged across all folds

1.1.2 Noisy Dataset

The resulting metrics for the performance of our decision tree on the noisy dataset are as follows:

Actual	Predicted			
	Class 1	Class 2	Class 3	Class 4
Class 1	45.4	1.5	1.7	1.4
Class 2	1.6	45.6	1.9	0.9
Class 3	1.3	2.0	45.8	0.9
Class 4	1.7	1.0	1.3	46.0

Confusion Matrix 2: Noisy dataset with no pruning

Accuracy: 0.914

Precision	Class 1	Class 2	Class 3	Class 4
	0.908	0.910	0.903	0.935
Recall	Class 1	Class 2	Class 3	Class 4
	0.908	0.912	0.916	0.920
F1-score	Class 1	Class 2	Class 3	Class 4
	0.908	0.911	0.909	0.927

1.2 Discussion

1.2.1 Result Analysis

The clean dataset shows excellent classification performance with very high accuracy – most classification errors (66%) occur between rooms 2 and 3, likely due to the fact that there is a WIFI hub on the wall separating them. When introducing noise, the model’s performance declines moderately, as expected, with errors distributed fairly uniformly across all classes, suggesting no particular room combinations were systematically misclassified.

1.2.2 Dataset Differences

Our results on the clean dataset were better, across all metrics, than on the noisy dataset. This is not surprising as decision trees learn hard splits based on the data.

Given a noisy dataset we’d expect the tree to overfit based on noise present in the training set and learn incorrect splits. For the clean dataset this noise isn’t present, hence the tree will only learn “true” splits which generalise better.

2 Pruning

2.1 Metrics

For the final step, we pruned the decision trees to maximise performance on the validation set and tested for generalisation on unseen data using nested cross-validation.²

²Results were averaged across all folds.

2.1.1 Clean Dataset

The resulting metrics for the performance of our decision tree on the clean dataset are as follows:

Actual	Predicted			
	Class 1	Class 2	Class 3	Class 4
Class 1	49.7	0.0	0.2	0.1
Class 2	0.0	48.5	1.5	0.0
Class 3	0.4	1.3	48.0	0.3
Class 4	0.3	0.0	0.2	49.5

Confusion Matrix 3: Clean dataset with pruning

Accuracy: 0.978

Precision	Class 1	Class 2	Class 3	Class 4
	0.986	0.974	0.962	0.992
Recall	Class 1	Class 2	Class 3	Class 4
	0.994	0.970	0.960	0.990
F1-score	Class 1	Class 2	Class 3	Class 4
	0.990	0.972	0.961	0.991

Average tree depth before pruning: 11.93

Average tree depth after pruning: 8.02

2.1.2 Noisy Dataset

The resulting metrics for the performance of our decision tree on the noisy dataset are as follows:

Actual	Predicted			
	Class 1	Class 2	Class 3	Class 4
Class 1	47.0	0.9	1.1	1.0
Class 2	1.0	47.1	1.3	0.6
Class 3	0.9	1.4	47.0	0.7
Class 4	1.1	0.7	0.9	47.3

Confusion Matrix 4: Noisy dataset with pruning

Accuracy: 0.942

Precision	Class 1	Class 2	Class 3	Class 4
	0.940	0.940	0.934	0.954
Recall	Class 1	Class 2	Class 3	Class 4
	0.940	0.942	0.940	0.946
F1-score	Class 1	Class 2	Class 3	Class 4
	0.940	0.941	0.937	0.950

Average tree depth before pruning: 18.17

Average tree depth after pruning: 14.02

2.2 Discussion

2.2.1 Result Analysis

On the clean dataset, pruning didn't improve the performance since pruning is intended to prevent decisions being made as a result of noise – which is not present in the clean dataset by definition. On the noisy data, the pruning algorithm improves all performance metrics (accuracy by 3%), a clear improvement made possible by removing the overfitted decisions based on noise in data. The performance gain was consistent across all labels³.

2.2.2 Depth Analysis

Trees trained on the noisy dataset were deeper both before and after pruning. While pruning reduced depth similarly (~ 4 nodes shallower) in both cases, it resulted in negligible performance change on clean data versus a clear improvement for noisy data. We speculate that on the clean dataset pruning removes unnecessarily fine-grained splits chasing perfect accuracy, while on the noisy dataset it eliminates harmful splits learned due to the noise.

³After removing the noise with pruning we again see that deciding between rooms 2 and 3 is most difficult

3 Tree Visualisation

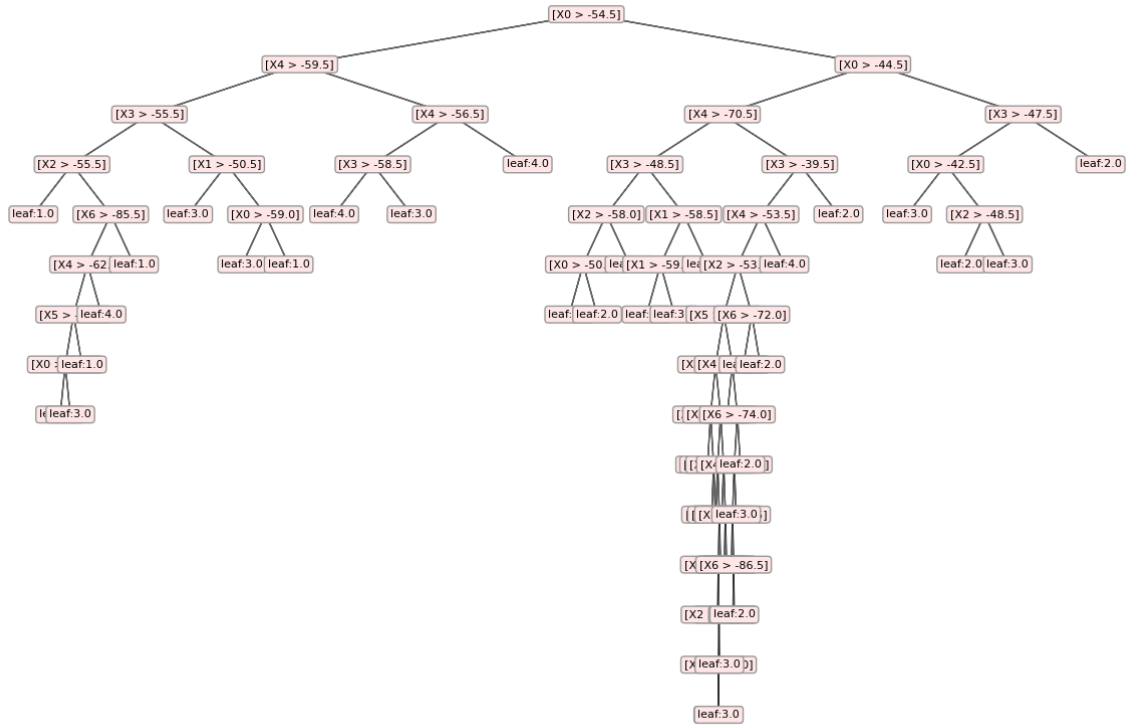


Figure 1: Visualisation of the decision tree for the clean dataset.